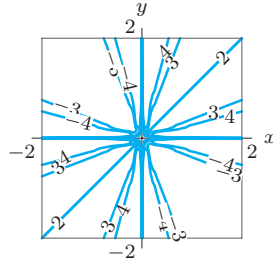
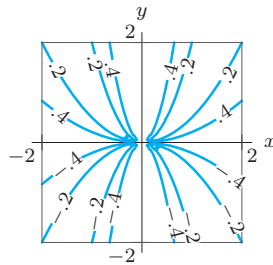


- 7 None
 9 (1, 2)
 11 (a)



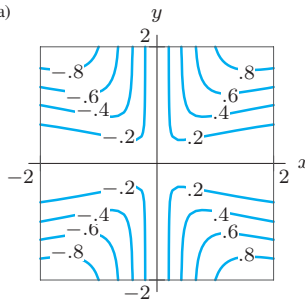
- (b) No
 (c) No
 (d) No
 (e) Exist, not continuous

13 (a)



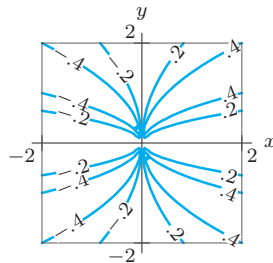
- (b) Yes
 (c) Yes
 (d) No
 (e) Exist, not continuous

15 (a)



- (b) Yes
 (d) No
 (f) No

17 (a)



- (c) No, no

- 19 (a) $f_x(x, y) = (x^4y + 4x^2y^3 - y^5)/(x^2 + y^2)^2$
 $f_y(x, y) = (x^5 - 4x^3y^2 - xy^4)/(x^2 + y^2)^2$
 (c) Yes
 (d) Yes

21 Counterexample: $\sqrt{x^2 + y^2}$

23 $f(x, y) = \sqrt{x^2 + y^2}$

- 25 (a) Differentiable
 (b) Not differentiable
 (c) Not differentiable
 (d) Differentiable

Chapter 14 Review

- 1 Vector; $3e^{-1}\vec{i} - \frac{1}{2}e^{-1}\vec{j}$
 3 Vector; $-(\sin x)e^y\vec{i} + (\cos x)e^y\vec{j} + \vec{k}$
 5 $f_x = 2xy + 3x^2 - 7y^6$
 $f_y = x^2 - 42xy^5$
 7 $\pi/\sqrt{lg}$
 9 $f_x = \frac{2xy^3}{(x^2+y^2)^2}$, $f_y = \frac{x^4-x^2y^2}{(x^2+y^2)^2}$
 11 $\partial f/\partial p = (1/q)e^{p/q}$
 $\partial f/\partial q = -(p/q^2)e^{p/q}$
 13 $f_N = c\alpha N^{\alpha-1}V^\beta$
 15 $x/(2\sqrt{\omega x} \cos^2(\sqrt{\omega x}))$
 17 $\frac{270x^3y^7 - 168x^2y^6 - 15xy^2 + 16y}{(15xy - 8)^2}$
 19 $\pi xy/\sqrt{2\pi xyw - 13x^7y^3v}$
 21 $\frac{7}{2} \left(\frac{w-1}{x^2yw - xy^3w^7} \right)^{-9/2}$
 $\left(\frac{x^2y + 6xy^3w^7 - 7xy^3w^6}{(w-1)^2} \right)$
 23 $-1/(4\pi L\sqrt{LC})$
 25 $u_{xx} = e^x \sin y$, $u_{yy} = -e^x \sin y$
 27 $f_{xy} = f_{yx} = 2 \cos(x - 2y)$
 29 $2x\vec{i} + (2y + 3y^2)\vec{j}$
 31 $-((1/x)\vec{i} + (1/y)\vec{j} + (1/z)\vec{k})/(xyz)$
 33 $\nabla z = 2x \cos(x^2 + y^2)\vec{i} + 2y \cos(x^2 + y^2)\vec{j}$
 35 $\cos(x^2 + y^2 + z^2)(2x\vec{i} + 2y\vec{j} + 2z\vec{k})$
 37 $-\frac{(t^2-2t+4)}{(2s\sqrt{s})}\vec{i} + \frac{(2t-2)}{\sqrt{s}}\vec{j}$
 39 $y[\cos(xy) - \sin(xy)]\vec{i} + x[\cos(xy) - \sin(xy)]\vec{j}$
 41 $2\vec{i} + \vec{k}$
 43 -1
 45 0
 47 $2/\sqrt{3}$
 49 $5\vec{i} + 4\vec{j} + 3\vec{k}$
 51 $-4x - 3y + 4z = 9$
 53 $x + y + z = 3$
 55 $\cos t \sin(\cos t) - \sin^2 t \cos(\cos t)$
 57 $100t^3$
 59 $3/t + 2t/(t^2 + 1)$
 61 $Q(x, y) = 2 + 6x + y + 6x^2 + 3xy$
 63 $Q(x, y) = 1 + (x - 3) - \frac{1}{2}(y - 5) - \frac{1}{2}(x - 3)^2 + \frac{1}{2}(x - 3)(y - 5) - \frac{1}{8}(y - 5)^2$
 65 (a) $2x - 4y + az = a - 2$

(b) $a = 2$

- 67 (a) Q, R
 (b) Q, P
 (c) P, Q, R, S
 (d) None

71 $F = 684$ newtons,
 $\partial F/\partial m = 9.77$ newtons/kg,
 $\partial F/\partial r = -0.000214$ newtons/meter

- 73 (a) 20 hours per day
 (b) 18.615 hours per day

- 75 (a) P, S
 (b) R, S
 (c) P, Q, R, S
 (d) None

77 0.3

79 0.8

81 0

- 83 (a) $-5\sqrt{2}/2$
 (b) $4\vec{i} + \vec{j}$

- 85 (a) 98.387 ft/mile
 (b) 295.161 ft/hour

- 87 (a) $-4e^{-81}^\circ\text{C}/\text{meter}$
 (b) $-40e^{-81}^\circ\text{C}/\text{sec}$
 (c) $\sqrt{932}e^{-81}^\circ\text{C}/\text{meter}$

89 $-2x\vec{i} - 2y\vec{j}$

91 Yes

- 95 (a) $F_u(x, y, 3)$
 (b) $F_w(3, y, x)$
 (c) $F_u(x, y, x) + F_w(x, y, x)$
 (d) $F_u(x, y, xy) + yF_w(x, y, xy)$

97 $dP \approx 47.6 dL + 17.8 dK$

101 (a) $-3\vec{i} + 4\vec{j} - \vec{k}$

(b) $-3\vec{i} + 4\vec{j}$

105 $15^\circ\text{C}/\text{minute}$

107 Approx 7.5 at (1.94, 1.08)

109 $x - y$

- 111 (a) Negative, positive,
 Up if positive, down if negative
 (b) $\pi < t < 2\pi$
 (c) $0 < x < 3\pi/2$ and
 $0 < t < \pi/2$ or $3\pi/2 < t < 5\pi/2$

- 113 (a) $A_0 + A_1 + 2A_2 + A_3 + 2A_4 + 4A_5 + (A_1 + 2A_3 + 2A_4)(x - 1) + (A_2 + A_4 + 4A_5)(y - 2) + B_1t + C_1t$
 (b) $A_1B_1 + 2A_3B_1 + 2A_4B_1 + A_2C_1 + A_4C_1 + 4A_5C_1$

$$115 \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} \frac{du}{dt} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} \frac{du}{dt} + \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} \frac{dv}{dt} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} \frac{dv}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

Section 15.1

- 1 (I) and (V) Local maximum, (II) and (VI) Local minimum, (III) and (IV) Saddle point
 3 (a) None
 (b) E, G
 (c) D, F
 5 Local minimum
 7 Local maximum
 9 Local max: (4, 2)
 11 Local max: (1, 5)
 13 Saddle point: (0, 0)
 Saddle point: (2, 0)
 Local min: (1, 0.25)
 15 Saddle pts: (1, -1), (-1, 1)
 Local max: (-1, -1)
 Local min: (1, 1)
 17 Local max: (-1, 0)

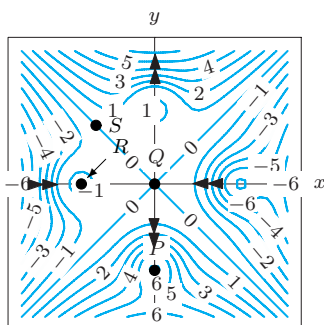
Saddle pts: $(1, 0), (-1, 4)$
Local min: $(1, 4)$

- 19 Saddle point: $(0, 0)$
Local max: $(1, 1), (-1, -1)$
21 $A = -2, B = 21$

- 23 (a) (a, b)
(b) $a = -1, b = 5$
(c) Local maximum

- 25 (a) Local maximum
(b) Saddle point
(c) Local minimum
(d) None of these

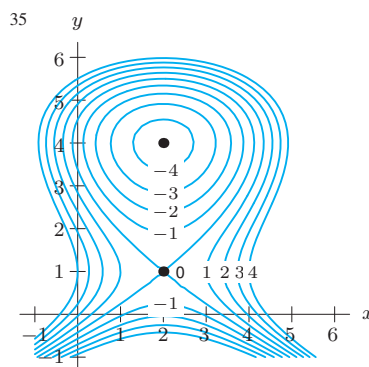
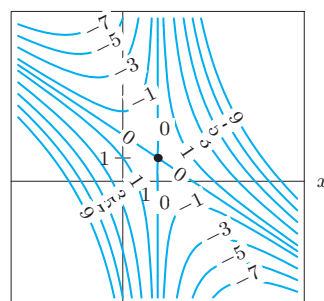
27



- 29 $(k\pi, l\pi)$,
for $k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$
 $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$,
for $k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$
 $(k\pi, l\pi), k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$ are saddle points
If k and l are both even or k and l are both odd,
then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local max
If k is even but l is odd or k is odd but l is even,
then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local min

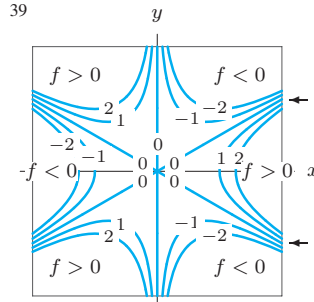
- 31 $y = 0, \pm 2\pi, \pm 4\pi, \dots$ Local minima

- 33 (a) (a, b) is a saddle point.
(b)



- 37 (a) $(0, 0)$
(b) $D = -24x^2$
(c) Saddle point

39



- 41 $(1, 3)$ could be saddle point
43 Can be saddle if f_{xy} large
45 $f(x, y) = 4 - (x - 2)^2 - (y + 3)^2$
47 False
49 True
51 False
53 True
55 False

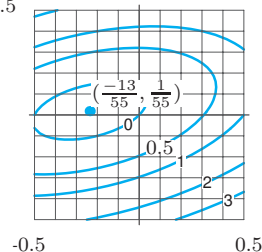
Section 15.2

- 1 Mississippi:
87 - 88 (max), 83 - 87 (min)
Alabama:
88 - 89 (max), 83 - 87 (min)
Pennsylvania:
89 - 90 (max), 80 (min)
New York:
81 - 84 (max), 74 - 76 (min)
California:
100 - 101 (max), 65 - 68 (min)
Arizona:
102 - 107 (max), 85 - 87 (min)
Massachusetts:
81 - 84 (max), 70 (min)
3 Max: 30.5 at $(0, 0)$
Min: 20.5 at $(2.5, 5)$
5 Min = 0 at $(0, 0)$
(not on boundary)
Max = 2 at $(1, 1), (1, -1),$
 $(-1, -1)$ and $(-1, 1)$
(on boundary)
7 max = 1 at $(1, 0)$ and $(-1, 0)$
(on boundary)
min = -1 at $(0, 1), (0, -1)$
(on boundary)

- 9 Global min

- 11 Global max = 0
No global min

- 13 (a) Local min: $(-13/55, 1/55)$
(b) 0.5



- 15 All edges $(32)^{1/3}$ cm
17 $l = w = h = 45$ cm
23 $q_1 = 300, q_2 = 225$.
25 (a) $L = \left[pA \left(\frac{a}{k} \right)^a \left(\frac{1}{b} \right)^{(a-1)} \right]^{1/(1-a-b)}$
 $K = \frac{1}{kb} L$
(b) No
27 $y = 24x^2/49 - 2/7$
31 (a) $p = \sqrt{P_0 P_F}$
(b) $p_1 = \sqrt[3]{P_0^2 P_F}, p_2 = \sqrt[3]{P_0 P_F^2}$
33 Must have global maximum if continuous and R closed, bounded
35 Local max not necessarily global max
37 $f(x, y) = x^2 + y^2, R$ is $0 \leq x, y \leq 1$
39 False
41 False
43 False
45 True

Section 15.3

- 1 Min = $-\sqrt{2}$, max = $\sqrt{2}$
3 Max: 20 at $(-1, 2)$;
Min: 0 at $(1, -2)$
5 Min = -22, max = 22
7 Global min: $1/2$
No global max
9 Min = $-\sqrt{35}$, max = $\sqrt{35}$
11 Max: 4 at $(2, 2, 1), (2, -2, -1),$
 $(-2, 2, -1), (-2, -2, 1)$;
Min: -4 at $(-2, -2, -1), (-2, 2, 1),$
 $(2, -2, 1), (2, 2, -1)$
13 Max = $f(\frac{1}{\sqrt{5}}, \frac{3}{\sqrt{5}}) = 2\sqrt{5}$
Min = $f(-\frac{1}{\sqrt{5}}, -\frac{3}{\sqrt{5}}) = -2\sqrt{5}$
15 No global extrema
17 Max: $30 + \sqrt{10}$ at $(0, -\sqrt{10})$;
Min: -5 at $(\pm 3, -1)$
19 1 at $(1/2, 1/2)$
21 0.5